# An untruncated linearity principle for Dedekind cuts forces the Cauchy and Dedekind reals to coincide

The Univalent Foundations Program

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*Editorial note.* The counted statement is the author's 'lem:untruncated-linearity-reals-coincide' (source lines 1803-1813, Lemma 11.4.1); the proof environment starts after one blank line, so the pairing is by adjacency.

*Editorial note.* The displayed assumption carries the internal label eq:untruncated-linearity, which is defined and used inside the retained statement and proof, so it needs no external entry.

*Editorial note.* The retained counter values (chapter 11, section 4, theorem offset 0) make the counted lemma print as Lemma 11.4.1, matching the original book numbering taken from the label-number dump of the pinned snapshot.

*Editorial note.* The single external reference is the weak-linearity equation of section 11.2; it is a dump-confirmed navigation pointer, kept as a pointer because a context from another section would print with the wrong section number.

*Editorial scope.* One original result of the HoTT Book, counted once: untruncated linearity forces the Cauchy and Dedekind reals to coincide (Lemma 11.4.1).

*Allowed standard prerequisites (editorial).*

- Dedekind reals and cuts. RD is the type of Dedekind cuts of rationals, with the usual order and inhabitation properties.. The assumption quantifies over x : RD and the proof uses inhabitedness of cuts at source line 1822.
- Embedding of RC into RD. The Cauchy reals embed into the Dedekind reals.. It reduces the goal to showing that every Dedekind real is merely a limit of a Cauchy sequence at source line 1818.
- Cauchy sequence of rationals. A Cauchy sequence of rationals converges to a real when the terms approximate it arbitrarily well.. It is the object constructed by the recursion in the proof.

*Source reference.* Original equation 11.2.3: sources/hott-reals/book-labelnumbers-first-edition.tex, source lines 1312–1312. Original result cited by the retained text but located elsewhere in the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned

book snapshot. It is kept as a pointer rather than as a context span because retaining it as context would either print it with the wrong section number or consume a number in the retained sequence.

**Lemma 11.4.1.** *If for every $x : \mathbb{R}_{\mathsf{d}}$ there merely exists*

$$c : \prod_{q,r:\mathbb{Q}} (q < r) \to (q < x) + (x < r) \tag{11.4.2}$$

*then the Cauchy and Dedekind reals coincide.*

*Proof.* Note that the type in (11.4.2) is an untruncated variant of (11.2.3), which states that $<$ is a weak linear order. We already know that $\mathbb{R}_{\mathsf{c}}$ embeds into $\mathbb{R}_{\mathsf{d}}$, so it suffices to show that every Dedekind real merely is the limit of a Cauchy sequence of rational numbers.

Consider any $x : \mathbb{R}_{\mathsf{d}}$. By assumption there merely exists $c$ as in the statement of the lemma, and by inhabitation of cuts there merely exist $a, b : \mathbb{Q}$ such that $a < x < b$. We construct a sequence $f : \mathbb{N} \to \{\, (q, r) \in \mathbb{Q} \times \mathbb{Q} \mid q < r \,\}$ by recursion:

(i) Set $f(0) :\equiv (a, b)$.

(ii) Suppose $f(n)$ is already defined as $(q_n, r_n)$ such that $q_n < r_n$. Define $s :\equiv (2q_n + r_n)/3$ and $t :\equiv (q_n + 2r_n)/3$. Then $c(s, t)$ decides between $s < x$ and $x < t$. If it decides $s < x$ then we set $f(n + 1) :\equiv (s, r_n)$, otherwise $f(n + 1) :\equiv (q_n, t)$.

Let us write $(q_n, r_n)$ for the $n$-th term of the sequence $f$. Then it is easy to see that $q_n < x < r_n$ and $|q_n - r_n| \leq (2/3)^n \cdot |q_0 - r_0|$ for all $n : \mathbb{N}$. Therefore $q_0, q_1, \ldots$ and $r_0, r_1, \ldots$ are both Cauchy sequences converging to the Dedekind cut $x$. We have shown that for every $x : \mathbb{R}_{\mathsf{d}}$ there merely exists a Cauchy sequence converging to $x$. □